

Classification of waste materials in very-high-resolution satellite imagery

State of the art and thesis proposal

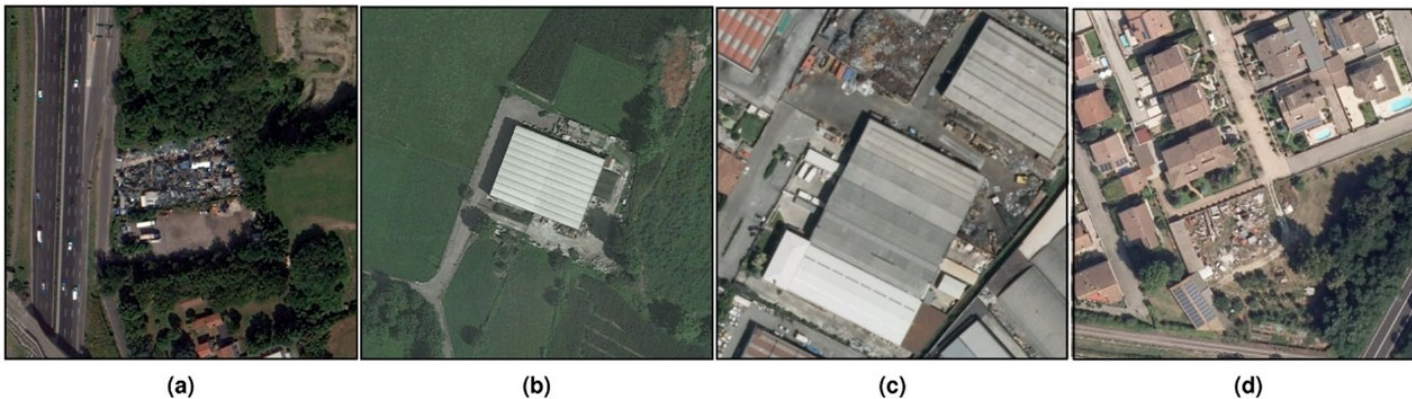
Alessandro Potenza · M.Sc. Computer Science and Engineering, AI · PERIVALLON

Context

Illegal waste dumping is an environmental crime with direct public-health consequences. Agencies (ARPA) have limited inspection capacity, and priority depends on what is dumped: rubble, plastics and asbestos-cement imply very different hazards.

Detecting dump sites from images is mature. Recognising the material is not: the reference survey (50 works, 1987-2023, almost all RGB) marks it as an open problem.

One thesis in the group has addressed material classification directly (Alari 2024). This work starts from there.



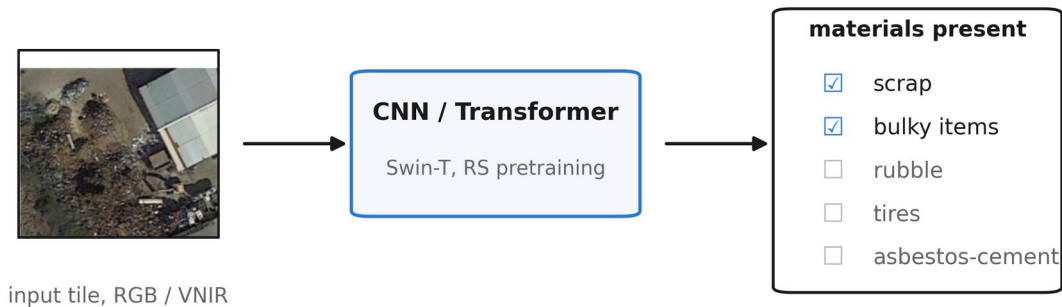
Task definition

Task: multi-label classification of waste materials in satellite images, at image level.

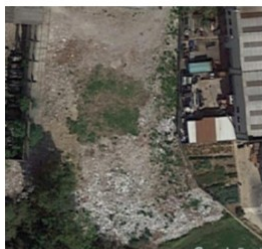
Input: VHR optical imagery, GSD 0.2 to 1.3 m (aerial RGB baseline; satellite VNIR arm, up to 8 bands). SWIR excluded: not in the planned acquisitions, and its 3.7 m GSD does not match the task. Sentinel-2 excluded: too coarse.

Technique: classification, not detection or segmentation. Annotations are image-level (Alari 2024: 11,477 multi-label); waste piles have no stable shape for boxes; no masks exist.

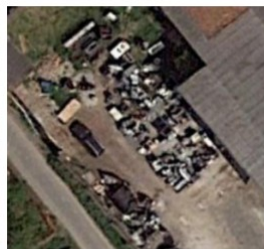
Research question: does VNIR input improve waste-material classification over RGB, and for which materials?



Materials: which are considered and why



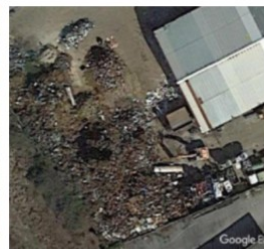
Rubble



Bulky items



Wood



Scrap

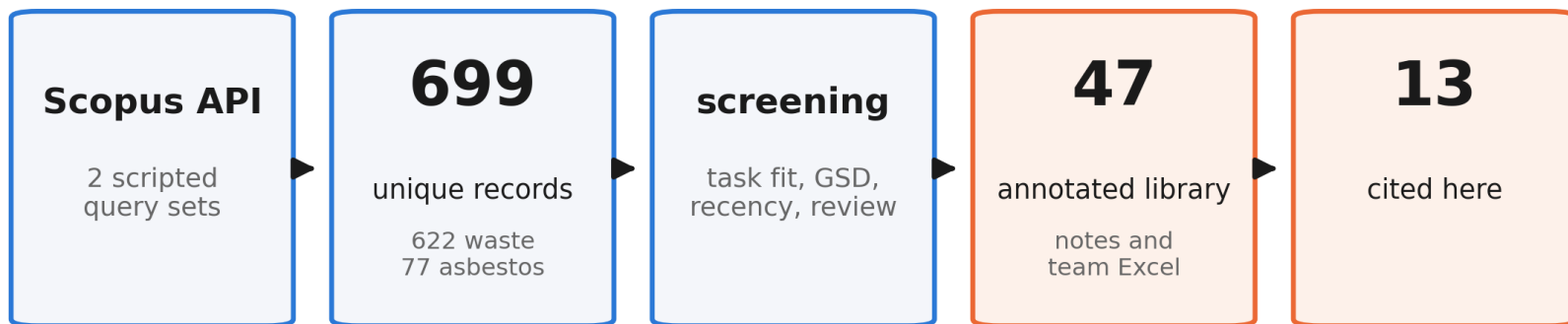


Plastic

Material	Target	Spectral analysis	Reason
Rubble, plastic, wood, tires	yes	yes	frequent and confusable in RGB
Asbestos-cement	yes	yes, dedicated pilot	public regional ground truth (WFS)
Vehicles, tanks, containers, scrap, bulky, big bags	yes	no	shape-based; RGB sufficient in literature
Sludge, foundry waste	yes	no	few labels; visually ambiguous at this GSD

Literature search: method and numbers

Scripted queries on the Scopus API (waste detection in remote sensing; asbestos roof mapping), deduplication, manual screening with explicit criteria, snowballing from the Fraternali 2024 survey and the Alari 2024 references.



Related work: site-level waste detection

Work (year)	Input / GSD	Task	Method	Result
Gibellini 2025	aerial RGB, 20 cm	classification	Swin-T, RSP pretraining	F1 92.0; cross-region -5.1
Disaitek 2024	Pléiades Neo, 30 cm	detection	operational service	~95% on sites >2 m2 (vendor)
CascadeDumpNet 2024	Pléiades, 50 cm	object detection	two-stage CNN + AutoML	mAP 84.6, cross-city transfer
Sun 2023	VHR RGB, 0.3-1 m	object detection	BCA-Net (Faster R-CNN)	~2,500 dumpsites, 28 cities
CWLD 2024	GF-2 + GE, 0.5-0.8 m	segmentation	DeepLabV3+ variant	F1 88.9, construction waste
AerialWaste, Torres 2023	aerial RGB, 20-50 cm	dataset + cls.	ResNet-FPN baseline	F1 80.7; 22 material tags

Related work: material-level classification

Work (year)	Input / GSD	Task	Result
Alari 2024 (PoliMi)	satellite RGB	multi-label cls.	wF1 69.2 (5 cat.), 59.4 (10 cat.)
Saba 2026	WV-3 VNIR, 1.24 m	pixel cls.	asbestos, Macro-F1 97.6
Bonifazi 2026	WV-3 VNIR+SWIR	pixel cls.	asbestos roofs, removal tracking
Abbasi 2024	aerial RGB	OBIA cls.	asbestos, OA ~96 (shape + time)
Cilia 2015	airborne HSI, 3 m	pixel cls.	asbestos, PA 89 / UA 86

One multi-material predecessor (Alari). The rest is asbestos only. Right: building-level asbestos classification on WorldView-3 (Bonifazi 2026).



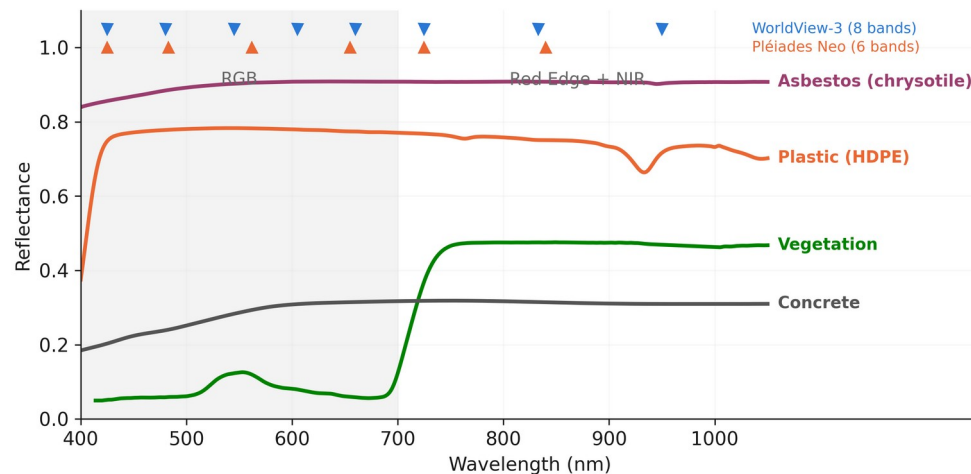
Where RGB falls short, and what VNIR adds

Different materials share the same colour at 0.3-1.3 μm : plastic sheets, asbestos-cement and concrete all appear grey.

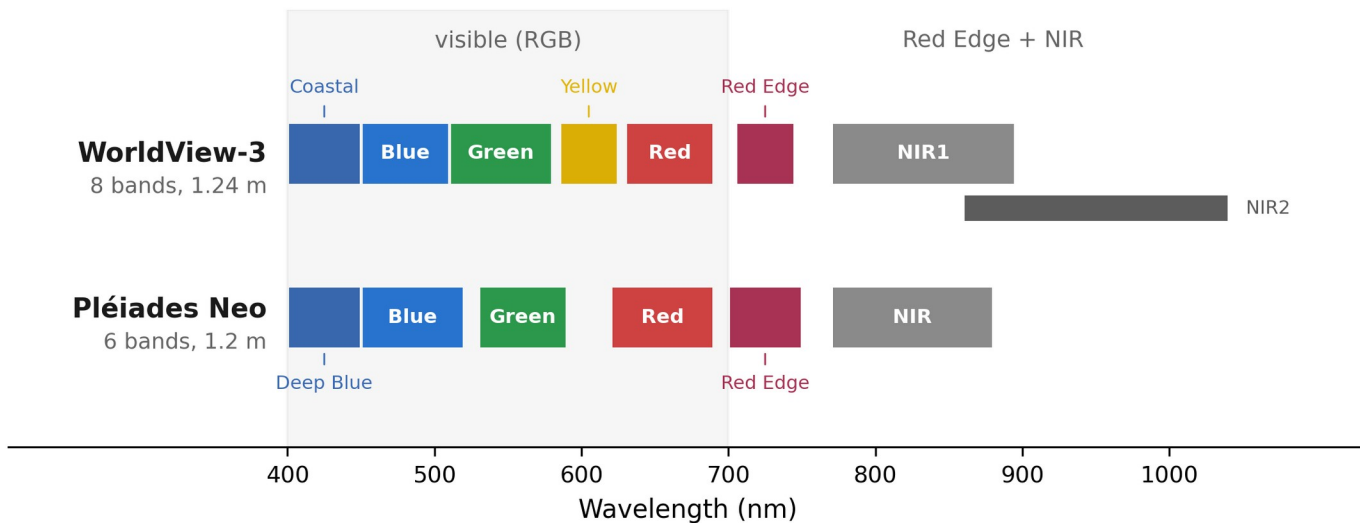
In the group predecessor, moving from 5 to 10 material categories costs 9.8 F1 points (69.2 to 59.4). Finer material distinctions are the hard part.

Red Edge and NIR separate vegetation, bare soil and weathered surfaces; in Saba 2026 they drive asbestos discrimination.

Whether this helps waste materials, and which ones, has not been measured. It is the object of this thesis.



Available imagery



Panchromatic: 0.31 m (WorldView-3), 0.30 m (Pléiades Neo). Access: commercial, or free quota via ESA proposal.

This is the band budget of the study: no SWIR, no Sentinel-2. RGB baseline runs on AerialWaste (20-50 cm).

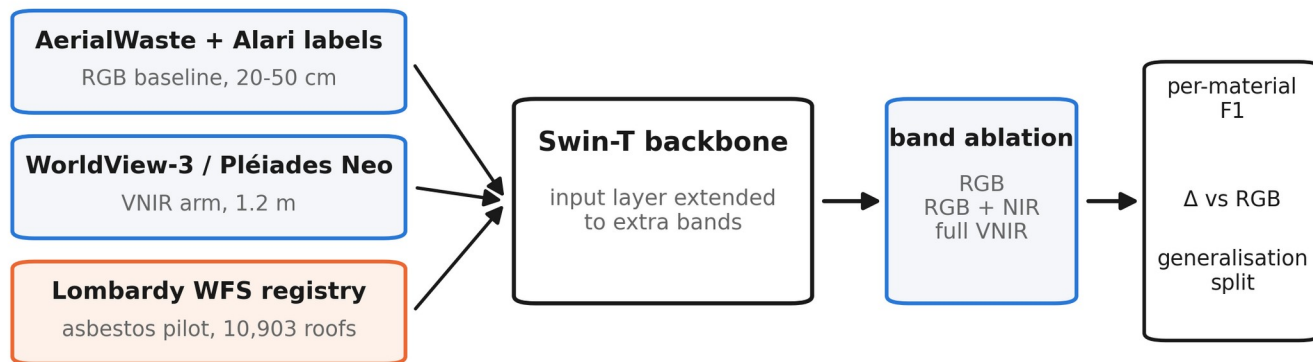
What is missing in the literature

1. Multi-material classification has one direct precedent, with ample margin: wF1 59-69 (Alari 2024).
2. No work measures the added value of VNIR bands over RGB for waste materials at very high resolution.
3. Results are reported as aggregate scores; per-material behaviour is not analysed.
4. Generalisation across regions is rarely evaluated. At site level it costs 5 F1 points (Gibellini 2025).
5. Asbestos is studied on roofs, in isolation, and never inside a waste-material taxonomy.

Proposed work: approach

Multi-label image classification, continuing the group line (Gibellini 2025, Alari 2024). Same taxonomy, input extended from RGB to VNIR.

Band ablation with everything else fixed: same architecture, same splits, only the input changes. Asbestos pilot first: public labels, one material, clear VNIR evidence.



Proposed work: evaluation

Per-material F1 alongside weighted and macro averages:
aggregates hide exactly the classes this thesis targets.

Delta versus the RGB baseline for each band configuration, with
confidence intervals over repeated runs.

Generalisation: train and test on disjoint geographic areas, besides
the standard split.

Reference points: wF1 69.2 / 59.4 (Alari 2024); Macro-F1 97.6
(Saba 2026, per-pixel) as upper reference for the pilot, not directly
comparable.

If VNIR does not help a material, that is a documented negative
result with practical value for sensor choice.

	standard split	cross-region split
RGB	baseline F1	baseline F1
RGB + NIR	$\Delta F1$ per material	$\Delta F1$ per material
full VNIR	$\Delta F1$ per material	$\Delta F1$ per material

References

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- Saba et al. 2026. J. Hazardous Materials. · Bonifazi et al. 2026. Geomatics. · Abbasi et al. 2024. RSASE. · Cilia et al. 2015. ISPRS IJGI.
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